

$$\begin{bmatrix} 0.707 & -0.707 & 0 \\ 1.250 & 1.250 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Chapter 4: Introduction to Matrices

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Matrices and Geometry

A square matrix can perform any *linear transformation*. What's that?

- Preserves straight lines
- Preserves parallel lines.
- No translation: the axes do not move.

Linear Transformations

- Rotation
- Scaling
- Orthographic projection
- Reflection
- Shearing

Axial Displacements

Can rewrite any vector $\mathbf{v} = [x \ y \ z]$ as a sum of *axial displacements*.

$$\begin{aligned} \mathbf{V} &= [x \ y \ z] \\ &= [x \ 0 \ 0] + [0 \ y \ 0] + [0 \ 0 \ z] \\ &= x [1 \ 0 \ 0] + y [0 \ 1 \ 0] + z [0 \ 0 \ 1] \end{aligned}$$

Basis Vectors

- Define three unit vectors along the axes:

$$\mathbf{p} = [1 \ 0 \ 0]$$

$$\mathbf{q} = [0 \ 1 \ 0]$$

$$\mathbf{r} = [0 \ 0 \ 1].$$

- Then we can rewrite the axial displacement equation as

$$\mathbf{v} = x\mathbf{p} + y\mathbf{q} + z\mathbf{r}$$

- $\mathbf{p}$, $\mathbf{q}$, $\mathbf{r}$ are known as *basis vectors*

Arbitrary Basis Vectors

- Can use any three linearly independent vectors

$$\mathbf{p} = [p_x \ p_y \ p_z]$$

$$\mathbf{q} = [q_x \ q_y \ q_z]$$

$$\mathbf{r} = [r_x \ r_y \ r_z]$$

- *Linearly independent* means that there do not exist scalars a, b, c such that:

$$a\mathbf{p} + b\mathbf{q} + c\mathbf{r} = \mathbf{0}$$

Orthonormal Basis Vectors

- Best to use an orthonormal basis
- Orthonormal means unit vectors that are pairwise orthogonal:

$$\mathbf{p} \cdot \mathbf{q} = \mathbf{q} \cdot \mathbf{r} = \mathbf{r} \cdot \mathbf{p} = 0$$

- Otherwise things can get weird.

Matrix From Basis Vectors

Construct a matrix **M** using **p**, **q**, **r** as the rows of the matrix:

$$\mathbf{M} = \begin{bmatrix} \mathbf{p} \\ \mathbf{q} \\ \mathbf{r} \end{bmatrix} = \begin{bmatrix} p_x & p_y & p_z \\ q_x & q_y & q_z \\ r_x & r_y & r_z \end{bmatrix}$$

What Does This Matrix Do?

$$\begin{aligned} \begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} \mathbf{p}_x & \mathbf{p}_y & \mathbf{p}_z \\ \mathbf{q}_x & \mathbf{q}_y & \mathbf{q}_z \\ \mathbf{r}_x & \mathbf{r}_y & \mathbf{r}_z \end{bmatrix} \\ = \begin{bmatrix} x\mathbf{p}_x + y\mathbf{q}_x + z\mathbf{r}_x & x\mathbf{p}_y + y\mathbf{q}_y + z\mathbf{r}_y & x\mathbf{p}_z + y\mathbf{q}_z + z\mathbf{r}_z \end{bmatrix} \\ = x\mathbf{p} + y\mathbf{q} + z\mathbf{r} \end{aligned}$$

Transformation by a Matrix

- If we interpret the rows of a matrix as the basis vectors of a coordinate space, then multiplication by the matrix performs a coordinate space transformation.
- If $\mathbf{aM} = \mathbf{b}$, we say that vector $\mathbf{a}$ is *transformed* by the matrix $\mathbf{M}$ into vector $\mathbf{b}$.

Conversely

See what **M** does to the original basis vectors $[1\ 0\ 0]$, $[0\ 1\ 0]$, $[0\ 0\ 1]$.

$$\begin{aligned} \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{bmatrix} &= \begin{bmatrix} m_{11} & m_{12} & m_{13} \end{bmatrix} \\ \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{bmatrix} &= \begin{bmatrix} m_{21} & m_{22} & m_{23} \end{bmatrix} \\ \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{bmatrix} &= \begin{bmatrix} m_{31} & m_{32} & m_{33} \end{bmatrix} \end{aligned}$$

Visualize The Matrix

- Each row of a matrix is a basis vector after transformation.
- Given an arbitrary matrix, visualize the transformation by its effect on the standard basis vectors – the rows of the matrix.
- Given an arbitrary linear transformation, create the matrix by visualizing what it does to the standard basis vectors and using that for the rows of the matrix.

2D Matrix Example

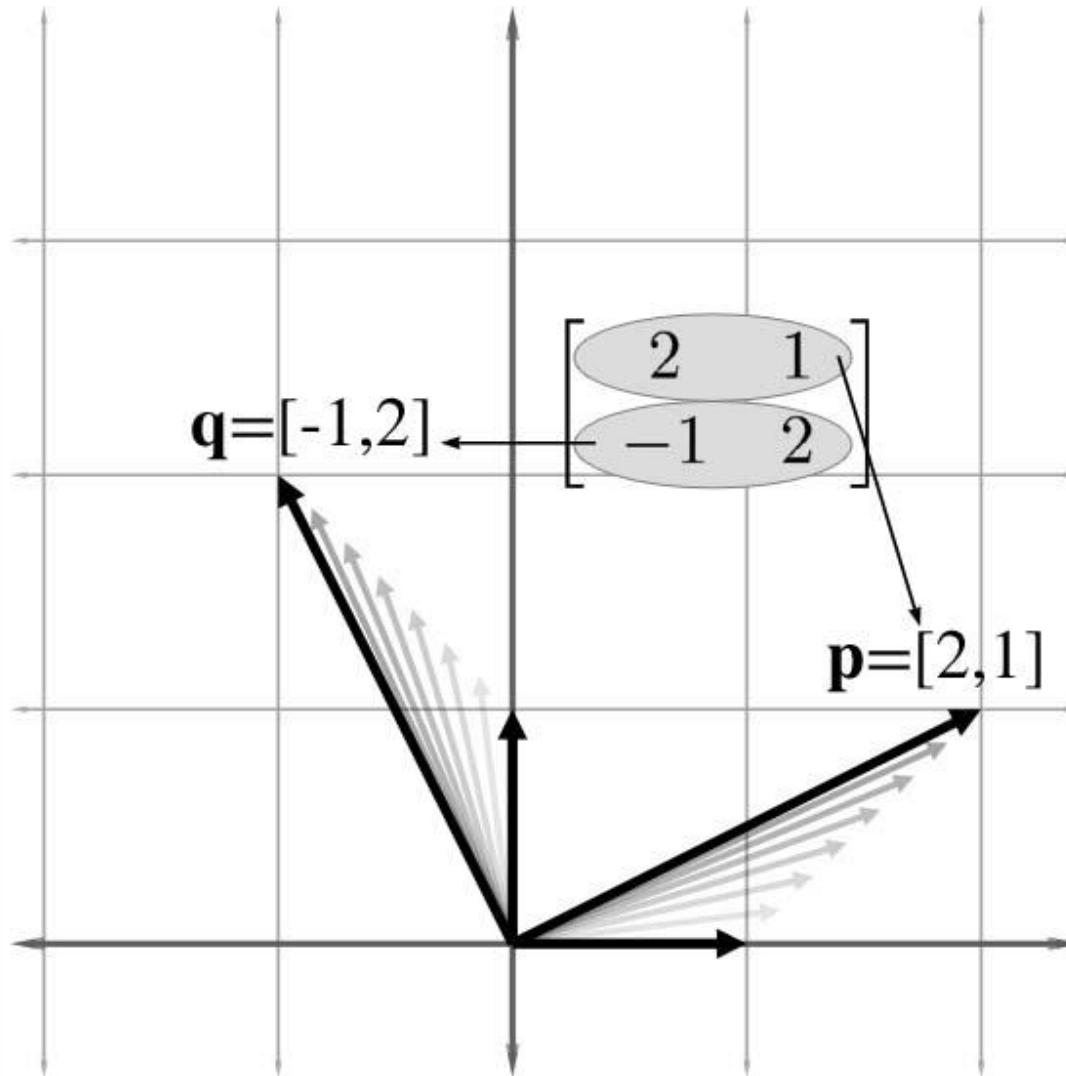
- What does the following 2D matrix do?

$$\mathbf{M} = \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}$$

- Extract the basis vectors (the rows of **M**)

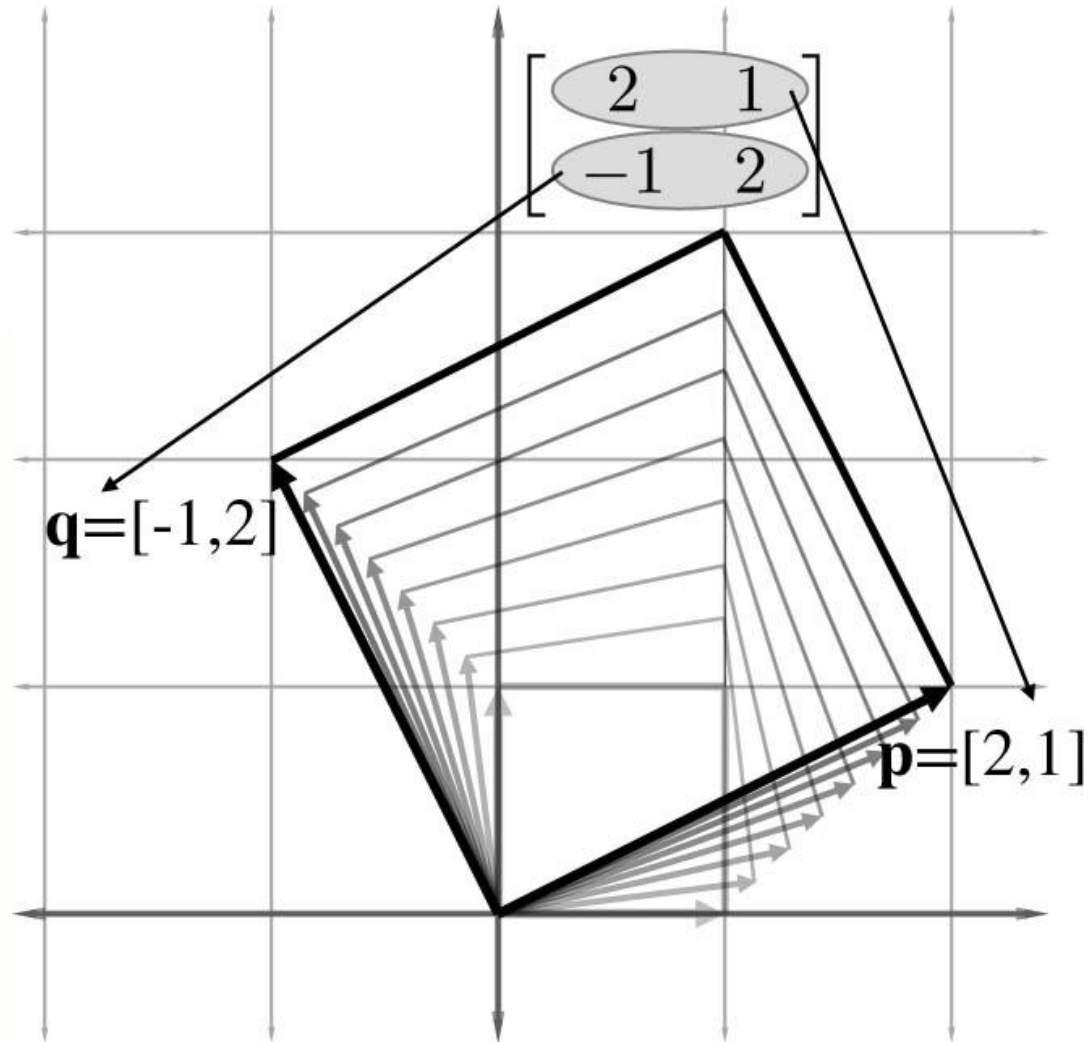
$$\mathbf{p} = [2, 1]$$

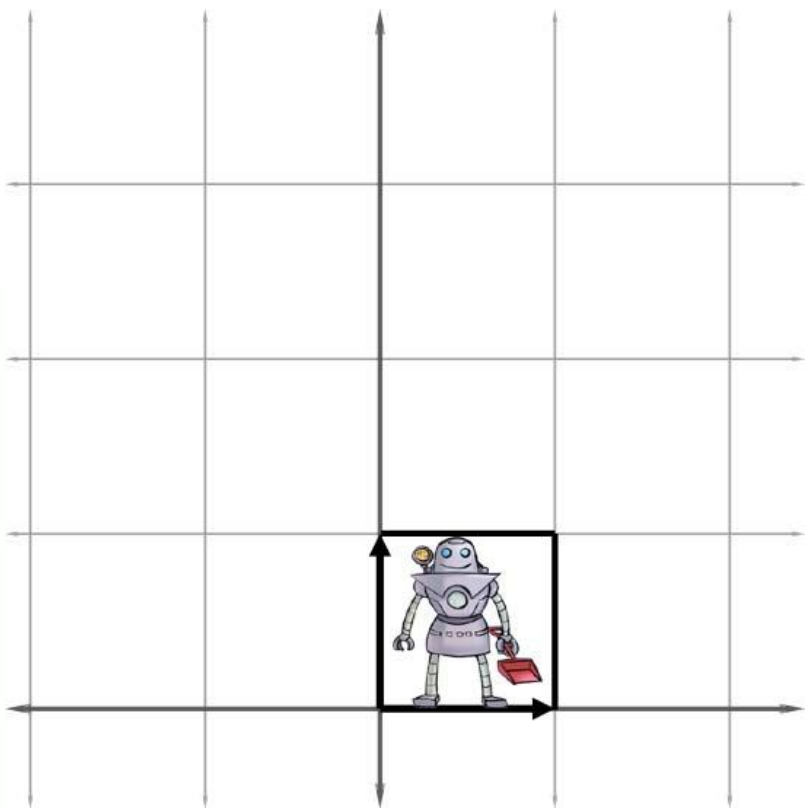
$$\mathbf{q} = [-1, 2]$$



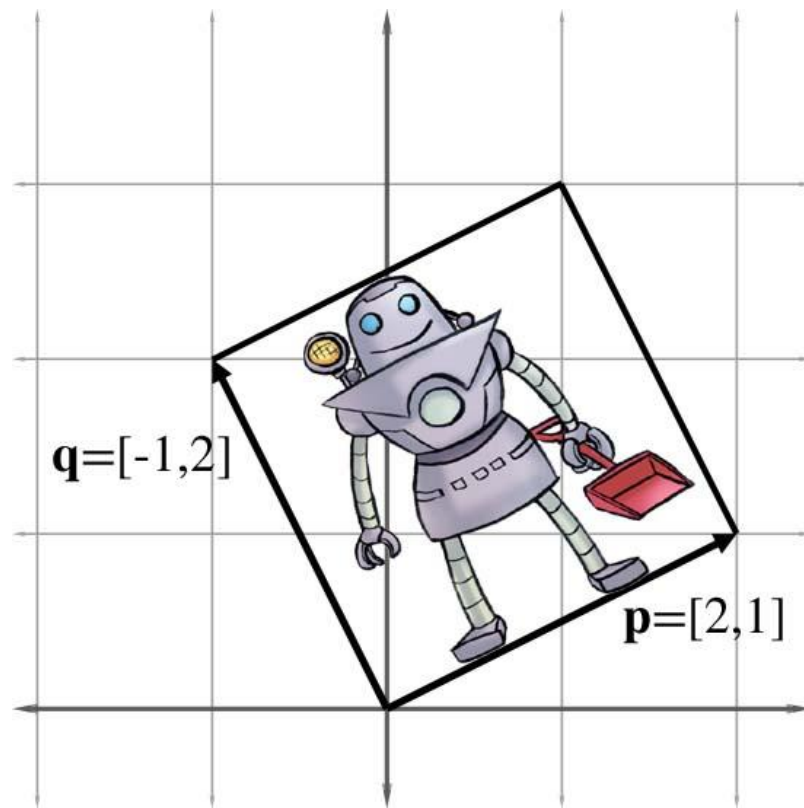
What's the Transformation?

- It moves the unit axes $[1, 0]$ and $[0, 1]$ to the new axes.
- It does the same thing to all vectors.
- Visualize a box being transformed from one coordinate system to the other.
- This is called a *skew box*.





Before

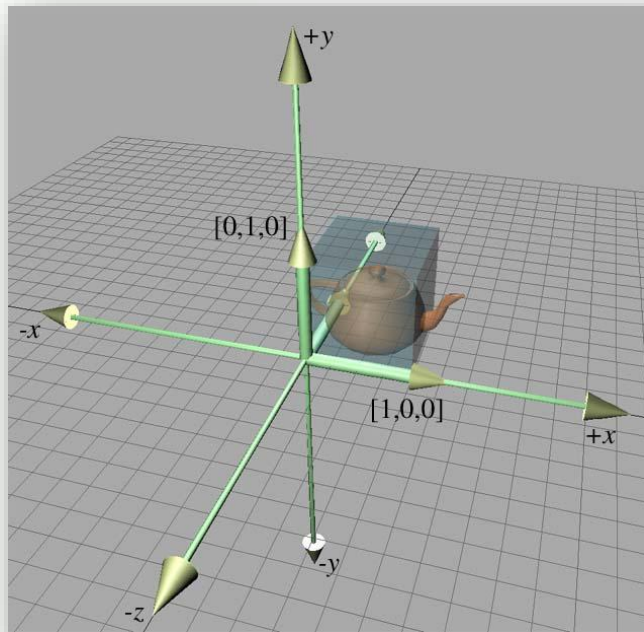


After

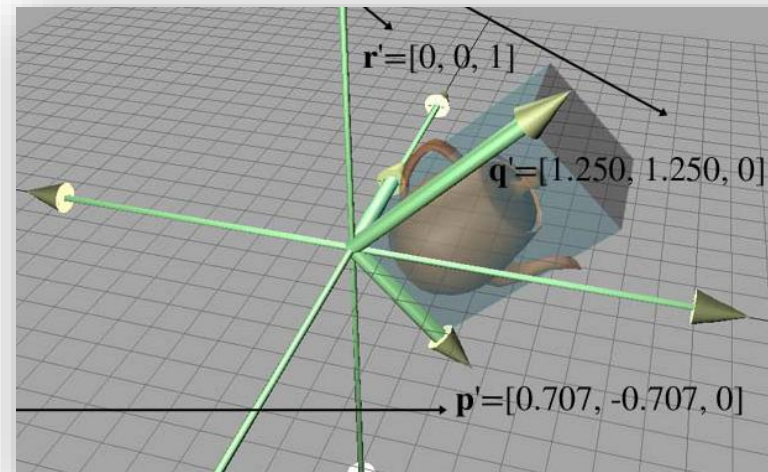
So What Does It Do?

- Rotates objects counterclockwise by a small amount.
- Scales them up by a factor of two.

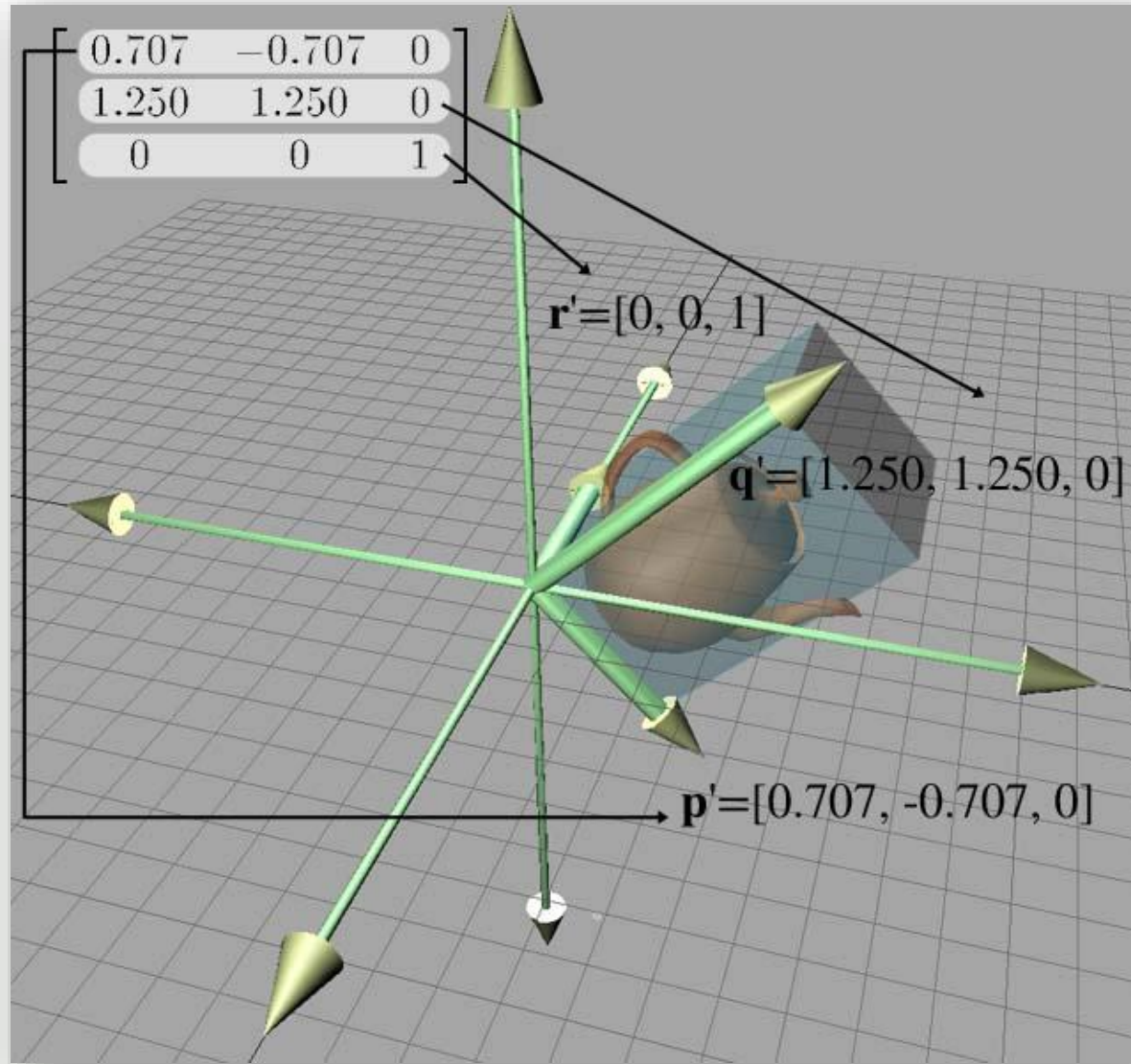
3D Transformation Example



Before



After



What's the Matrix?

Get rows of matrix from new basis vectors.

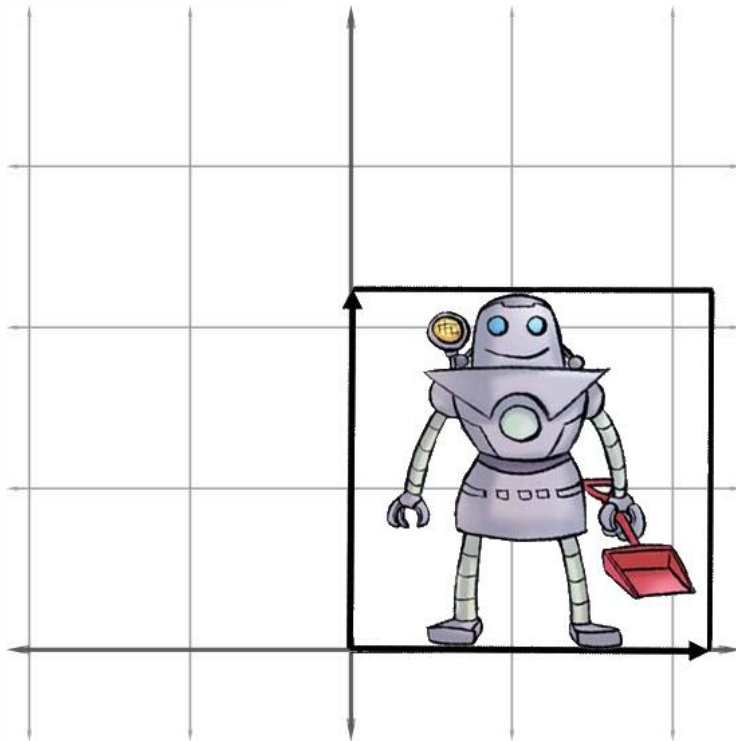
$$\begin{bmatrix} 0.707 & -0.707 & 0 \\ 1.250 & 1.250 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

So what does it do?

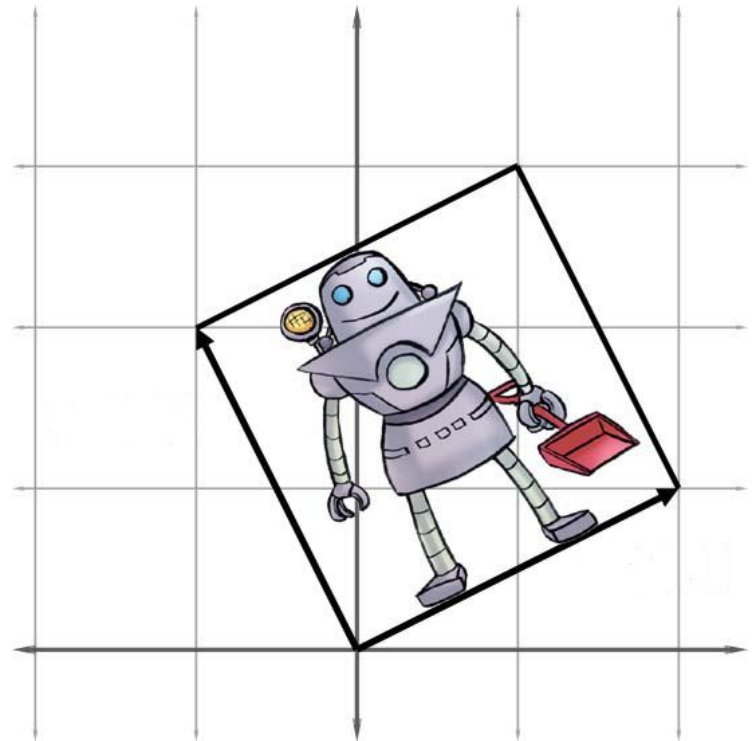
- Rotates objects clockwise by 45° .
- Scales them up along the y axis.

Section 5.1: Rotation

2D Rotation Around Point

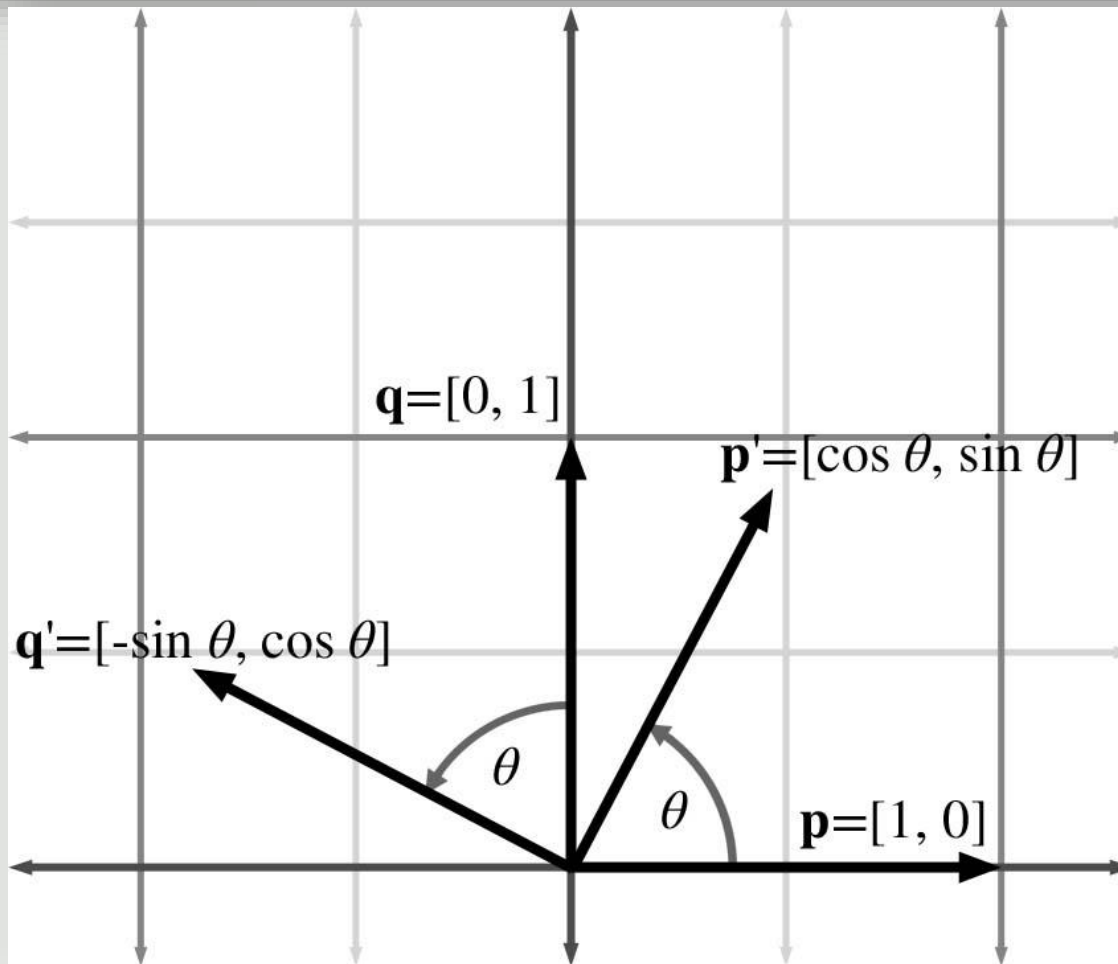


Before



After

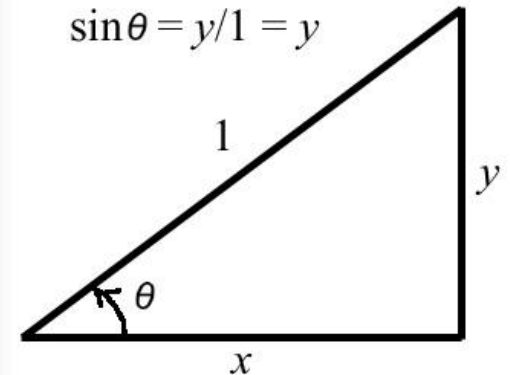
It's All About Rotating Basis Vectors!



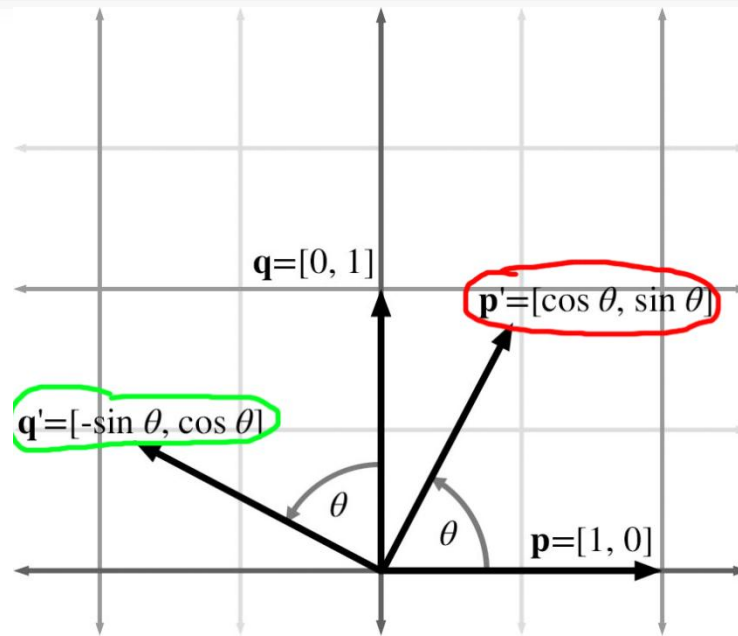
"SohCahToa"

$$\cos \theta = x/1 = x$$

$$\sin \theta = y/1 = y$$



Construct Matrix from Basis Vectors

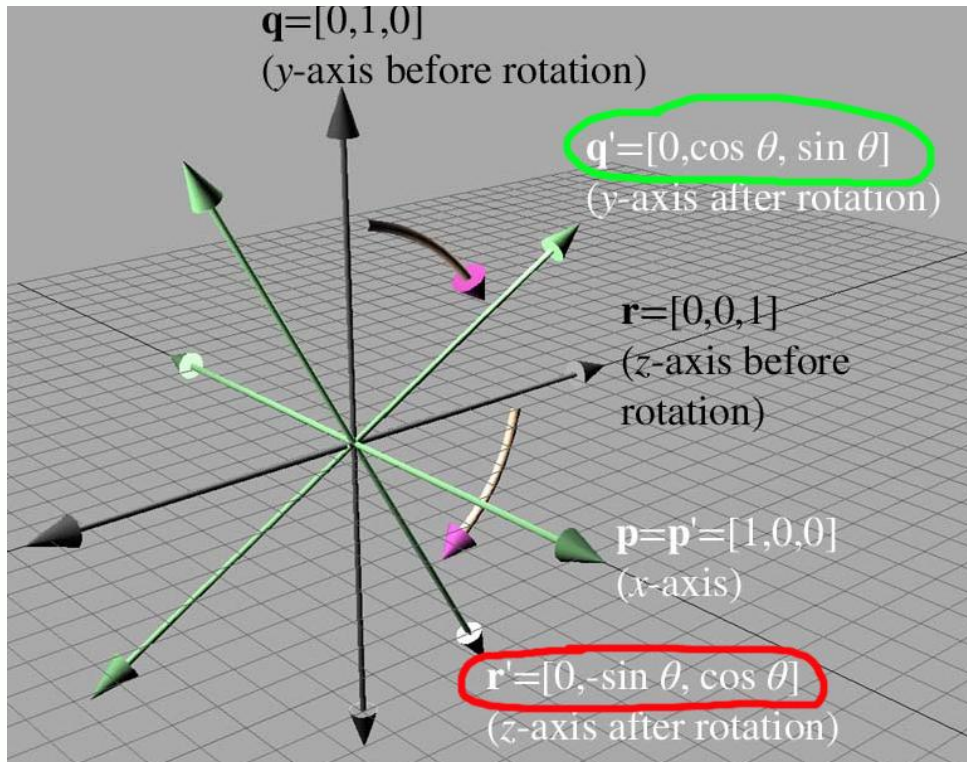


$$\mathbf{R}(\theta) = \begin{bmatrix} -\mathbf{p}' & -\mathbf{q}' \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

3D Rotation About Cardinal Axis

- In 3D, rotation occurs about an axis rather than a point as in 2D.
- The most common type of rotation is a simple rotation about one of the cardinal axes.
- We'll need to establish which direction of rotation is “positive” and which is “negative.”
- We're going to obey the left-hand rule for this (review Chapter 1).

3D Rotate About x-axis



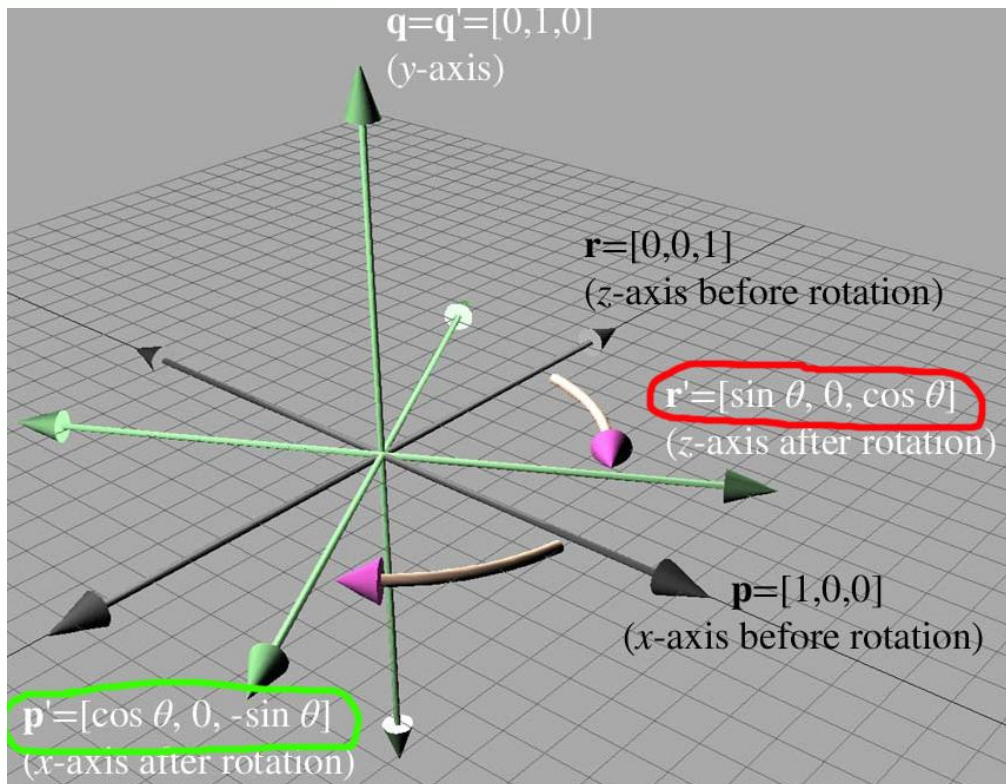
$$\mathbf{R}_x(\theta) = \begin{bmatrix} -\mathbf{p}' \\ -\mathbf{q}' \\ -\mathbf{r}' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{bmatrix}$$

Compare to 2D Case

$$\mathbf{R}(\theta) = \begin{bmatrix} -\mathbf{p}' \\ -\mathbf{q}' \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

$$\mathbf{R}_x(\theta) = \begin{bmatrix} -\mathbf{p}' \\ -\mathbf{q}' \\ -\mathbf{r}' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{bmatrix}$$

3D Rotate About y-axis



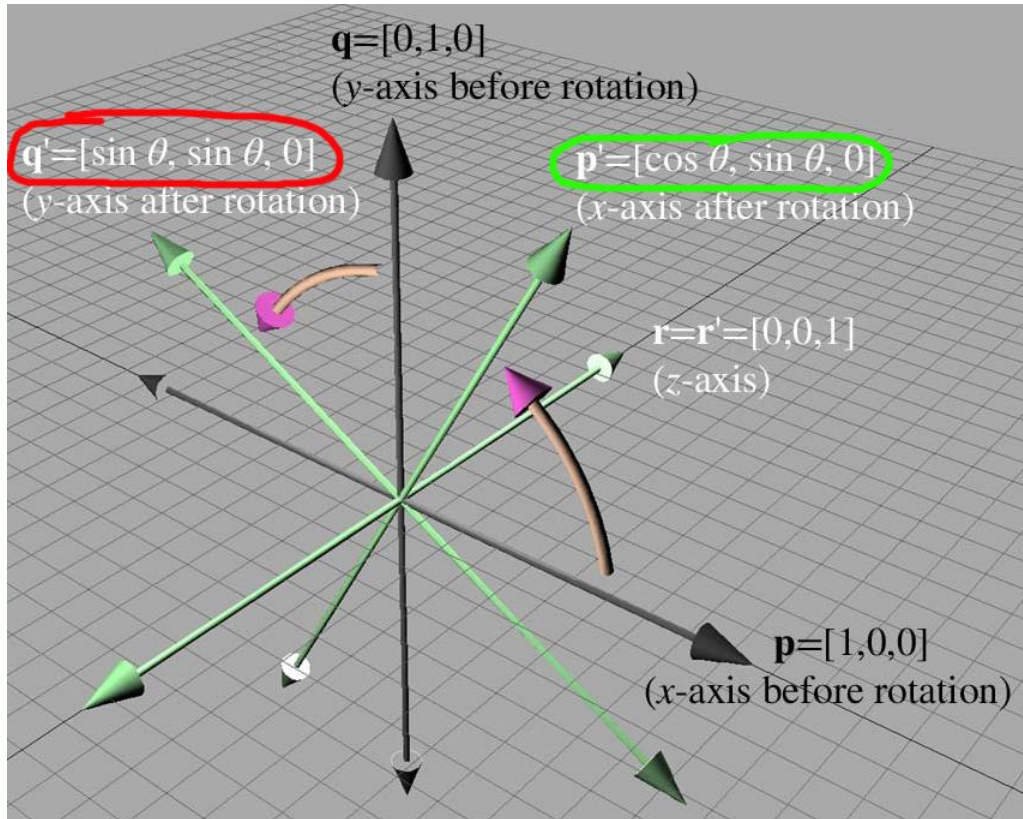
$$\mathbf{R}_y(\theta) = \begin{bmatrix} -\mathbf{p}' \\ -\mathbf{q}' \\ -\mathbf{r}' \end{bmatrix} = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix}$$

Compare to 2D Case

$$\mathbf{R}(\theta) = \begin{bmatrix} -\mathbf{p}' \\ -\mathbf{q}' \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

$$\mathbf{R}_y(\theta) = \begin{bmatrix} -\mathbf{p}' \\ -\mathbf{q}' \\ -\mathbf{r}' \end{bmatrix} = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix}$$

3D Rotate About z-axis



$$\mathbf{R}_z(\theta) = \begin{bmatrix} -\mathbf{p}' \\ -\mathbf{q}' \\ -\mathbf{r}' \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Compare to 2D Case

$$\mathbf{R}(\theta) = \begin{bmatrix} -\mathbf{p}' \\ -\mathbf{q}' \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

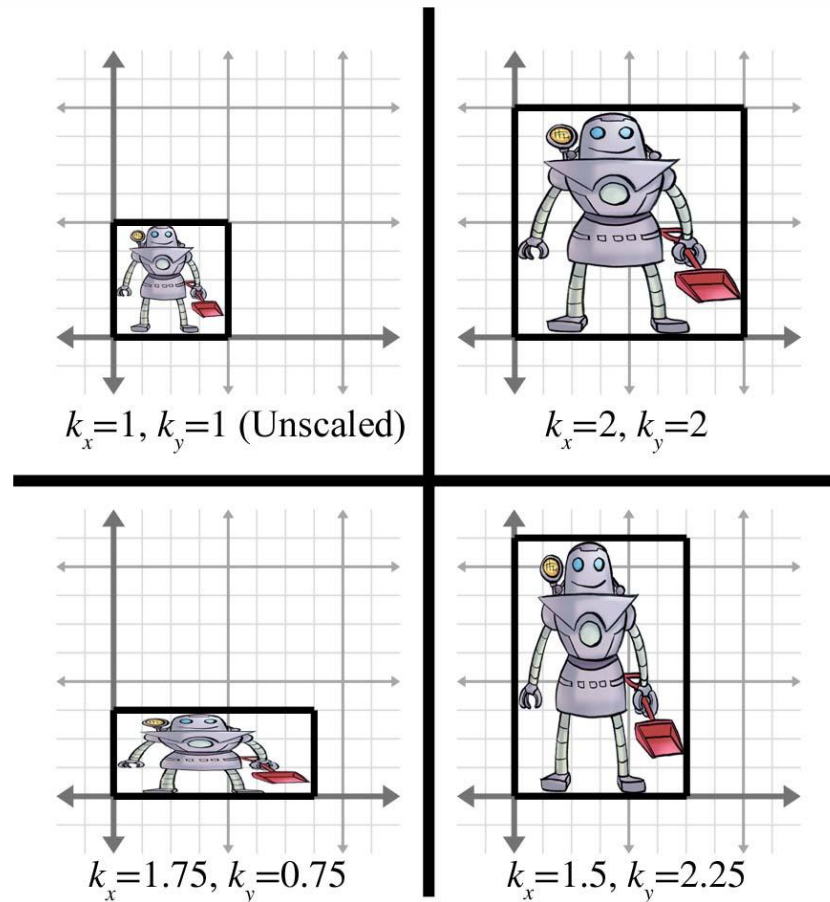
$$\mathbf{R}_z(\theta) = \begin{bmatrix} -\mathbf{p}' \\ -\mathbf{q}' \\ -\mathbf{r}' \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

3D Rotation About Noncardinal Axis

- We can also rotate about an arbitrary axis that passes through the origin.
- This is more complicated and less common than rotating about a cardinal axis.
- Game programmers worry less about this because rotation about an arbitrary axis can be expressed as a sum of rotations about cardinal axes (Euler).
- Details are left to the book.

Section 5.2: Scale

Scaling Along Cardinal Axes in 2D



Basis Vectors for Scale

The basis vectors **p** and **q** are independently affected by the corresponding scale factors:

$$\begin{aligned} \mathbf{p}' &= k_x \mathbf{p} = k_x \begin{bmatrix} 1 & 0 \end{bmatrix} = \begin{bmatrix} k_x & 0 \end{bmatrix} \\ \mathbf{q}' &= k_y \mathbf{q} = k_y \begin{bmatrix} 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & k_y \end{bmatrix} \end{aligned}$$

2D Scale Matrix

Constructing the 2D scale matrix $\mathbf{S}(k_x, k_y)$ from these basis vectors:

$$\mathbf{S}(k_x, k_y) = \begin{bmatrix} -\mathbf{p}' & - \\ -\mathbf{q}' & - \end{bmatrix} = \begin{bmatrix} k_x & 0 \\ 0 & k_y \end{bmatrix}$$

3D Scale Matrix

$$\mathbf{S}(k_x, k_y, k_z) = \begin{bmatrix} k_x & 0 & 0 \\ 0 & k_y & 0 \\ 0 & 0 & k_z \end{bmatrix}$$

$$\begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} k_x & 0 & 0 \\ 0 & k_y & 0 \\ 0 & 0 & k_z \end{bmatrix} = \begin{bmatrix} k_x x & k_y y & k_z z \end{bmatrix}$$

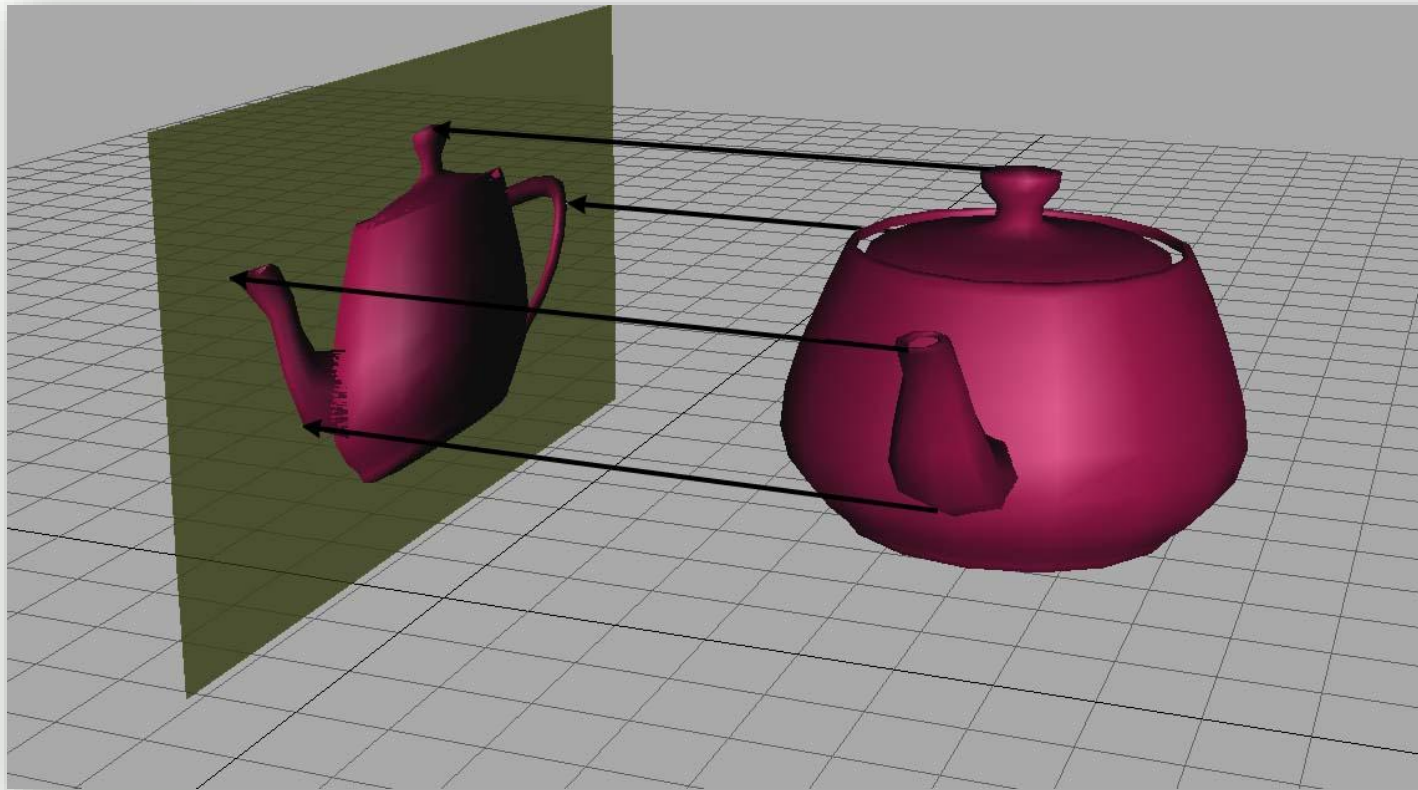
Scale in Arbitrary Direction

- The math for scaling in an arbitrary direction is intricate, but not too tricky.
- For game programmers it is not used very often.
- Details again left to the book.

Section 5.3:

Orthographic Projection

Orthographic Projection



Projecting Onto a Cardinal Axis

- Projection onto a cardinal axis or plane most frequently occurs not by actual transformation, but by simply discarding one of the dimensions while assigning the data into a variable of lesser dimension.
- For example, we may turn a 3D object into a 2D object by discarding the z components of the points and copying only x and y .
- However, we can also project onto a cardinal axis or plane by using a scale value of zero on the perpendicular axis.
- For completeness, we present the matrices for these transformations:

$$\mathbf{P}_x = \mathbf{S}([0 \ 1], 0) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\mathbf{P}_y = \mathbf{S}([1 \ 0], 0) = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

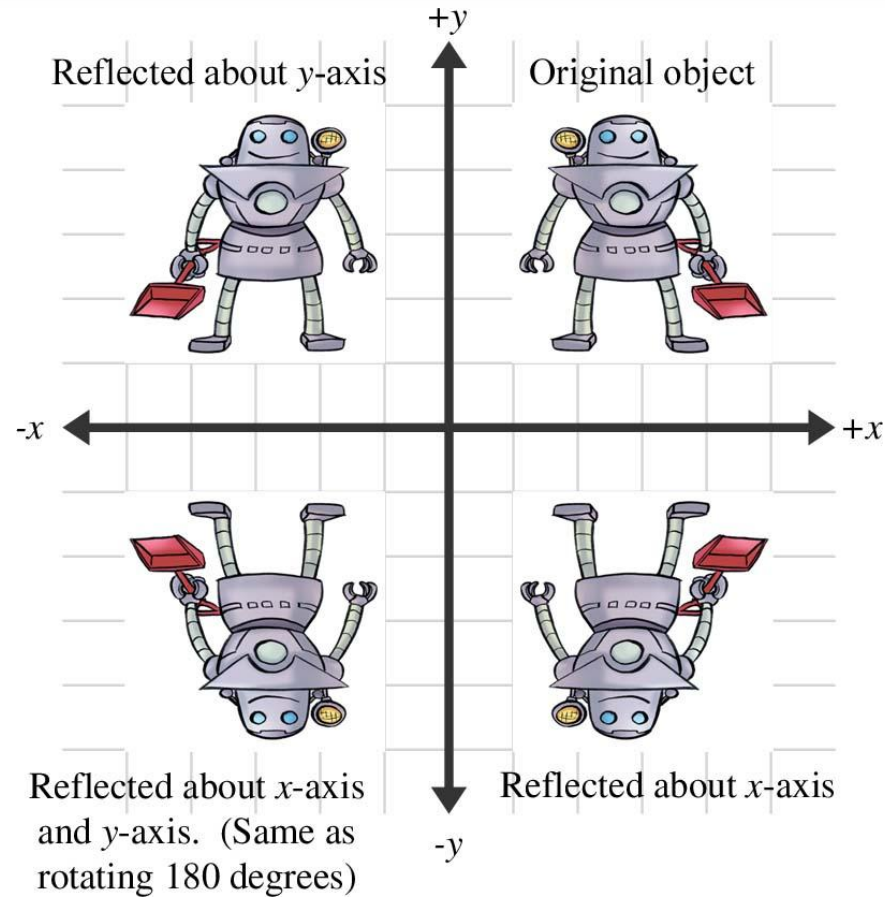
$$\mathbf{P}_{xy} = \mathbf{S}([0 \ 0 \ 1], 0) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{P}_{xz} = \mathbf{S}([0 \ 1 \ 0], 0) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{P}_{yz} = \mathbf{S}([1 \ 0 \ 0], 0) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

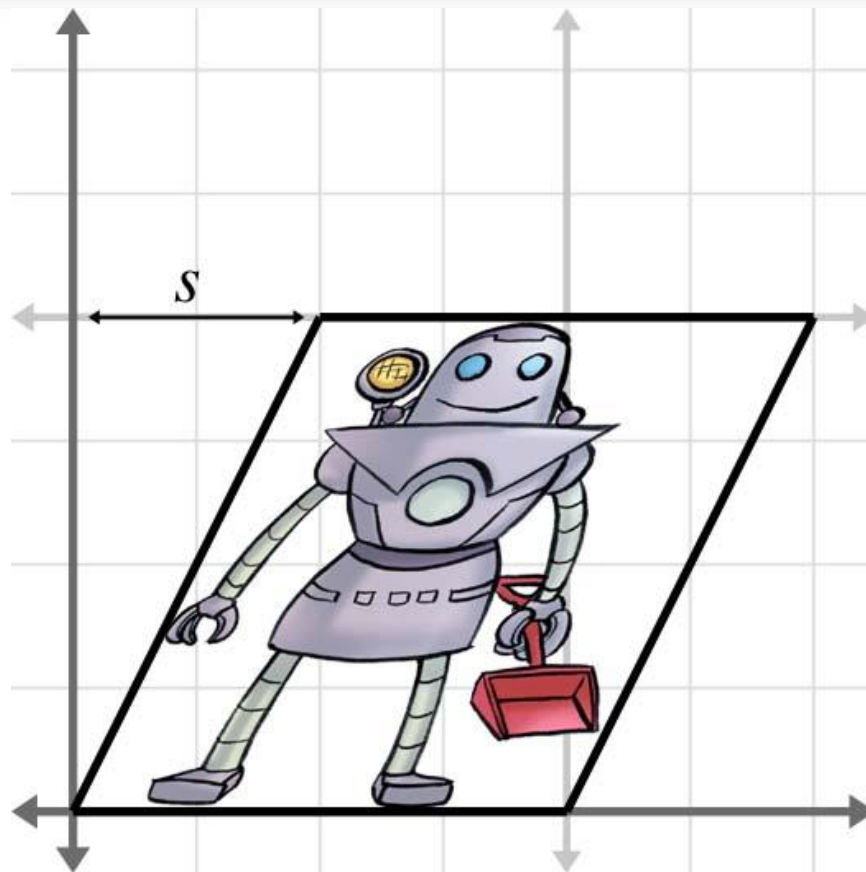
Section 5.4: Reflection

Reflection



Section 5.5: Shearing

Shearing



Shearing in 2D

- Shearing is a transformation that skews the coordinate space, stretching it non-uniformly.
- Angles are not preserved; however, surprisingly, areas and volumes are.
- The basic idea is to add a multiple of one coordinate to the other.
- For example, in 2D, we might take a multiple of y and add it to x , so that $x' = x + sy$.

2D Shear Matrices

Let $\mathbf{H}_x(s)$ be the shear matrix that shears the x coordinate by the other coordinate, y , by amount s .

$$\mathbf{H}_x(s) = \begin{bmatrix} 1 & 0 \\ s & 1 \end{bmatrix},$$

$$\mathbf{H}_y(s) = \begin{bmatrix} 1 & s \\ 0 & 1 \end{bmatrix}.$$

3D Shear Matrices

- In 3D, we can take one coordinate and add different multiples of that coordinate to the other two coordinates.
- The notation H_{xy} indicates that the x and y coordinates are shifted by the other coordinate, z .

3D Shear Matrices

$$\mathbf{H}_{xy}(s, t) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ s & t & 1 \end{bmatrix}$$

$$\mathbf{H}_{xz}(s, t) = \begin{bmatrix} 1 & 0 & 0 \\ s & 1 & t \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{H}_{yz}(s, t) = \begin{bmatrix} 1 & s & t \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Section 5.6:

Combining Transformations